

MATHEMATICS-II

Unit-IV: Ring Theory

Comprehensive Study Notes

Topic 1: Definition of Ring

1.1 What is a Ring?

A **ring** is an algebraic structure consisting of a set R equipped with two binary operations, typically called **addition** and **multiplication**, that generalize the arithmetic of integers.

1.2 Formal Definition

A non-empty set R together with two binary operations $+$ (addition) and $\cdot$ (multiplication) is called a **ring** if it satisfies the following axioms:

Axioms for Addition

Axiom	Description	Mathematical Form
Closure	Sum of two elements is in R	For all $a, b \in R$, $a+b \in R$
Associativity	Addition is associative	$(a+b)+c = a+(b+c)$
Commutativity	Addition is commutative	$a+b = b+a$
Identity	Additive identity (0) exists	There exists $0 \in R$ such that $a+0 = 0+a = a$
Inverses	Each element has additive inverse	For each a , there exists $-a$ such that $a+(-a)=0$

Axioms for Multiplication

Axiom	Description	Mathematical Form
Closure	Product of two elements is in R	For all $a, b \in R$, $ab \in R$

Associativity	Multiplication is associative	$(ab)c = a(bc)$
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Distributive Laws

Axiom	Description	Mathematical Form
Left Distributive	Multiplication distributes over addition	$a(b+c) = ab + ac$
Right Distributive	Multiplication distributes over addition	$(a+b)c = ac + bc$

Ring Axioms: $(R,+)$ is an Abelian Group + Multiplication Closure + Associativity + Distributivity

1.3 Key Points to Remember

1. A ring is an **abelian group** under addition
2. Multiplication is **not necessarily commutative**
3. **Multiplicative inverse** is **not required** (unlike fields)
4. A ring may or may not have a multiplicative identity (unity)

Topic 2: Types of Rings

2.1 Commutative Ring

A ring R is called **commutative** if multiplication is commutative:

$$ab = ba \text{ for all } a, b \in R$$

Example: $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ with usual addition and multiplication are commutative rings.

2.2 Ring with Unity

A ring R is called a **ring with unity** if there exists a **multiplicative identity** (denoted by 1) such that:

$$a \cdot 1 = 1 \cdot a = a \text{ for all } a \in R$$

Examples:

- $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are commutative rings with unity
- $\mathbb{Z}_n$ is a finite commutative ring with unity
- $M_{n \times n}(R)$ (matrices) is a noncommutative ring with unity

Examples without unity: $n\mathbb{Z}$ (multiples of n) is a commutative ring without unity

Note: The only ring where $1 = 0$ is the **zero ring** $R = \{0\}$.

2.3 Summary Table

Type	Condition	Example
Commutative Ring	$ab = ba$	$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
Non-commutative Ring	$ab \neq ba$	$M_{n \times n}(\mathbb{R})$

Ring with Unity	Has multiplicative identity 1	$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
Ring without Unity	No multiplicative identity	$n\mathbb{Z}$

Topic 3: Ring of Polynomials

3.1 Definition

If R is a commutative ring with unity, then the set of all polynomials with coefficients in R is denoted by:

$$R[x] = \{ a_0 + a_1x + a_2x^2 + \dots + a_nx^n \mid a_i \in R, n \geq 0 \}$$

This is called the **ring of polynomials** over R .

3.2 Properties

Property	Description
Ring	$R[x]$ is a commutative ring with unity
Unity	The polynomial 1 (constant polynomial) is the multiplicative identity
Examples	$\mathbb{Z}[x]$, $\mathbb{Q}[x]$, $\mathbb{R}[x]$, $\mathbb{C}[x]$ are commutative rings with unity
Degree	If $a_n \neq 0$, largest such integer is the degree

3.3 Examples

- **Example 1:** $\mathbb{Z}[x]$ – polynomials with integer coefficients
- **Example 2:** $\mathbb{Q}[x]$ – polynomials with rational coefficients
- **Example 3:** $\mathbb{R}[x]$ – polynomials with real coefficients
- **Example 4:** $\mathbb{Z}_n[x]$ – polynomials with coefficients modulo n

Topic 4: Ring of Functions

4.1 Definition

Let X be a set and R be a ring. The set of all functions from X to R forms a ring under pointwise addition and multiplication.

4.2 Operations

For functions $f, g: X \rightarrow R$:

Operation	Definition
Addition	$(f+g)(x) = f(x) + g(x)$
Multiplication	$(fg)(x) = f(x)g(x)$

4.3 Properties

- The set of all functions $X \rightarrow R$ forms a ring
- It is commutative if and only if R is commutative
- The zero function is the additive identity
- If R has unity, the constant function $f(x) = 1$ is the multiplicative identity

Special Case: The set of all continuous real-valued functions on the real line forms a commutative ring with unity.

Topic 5: Elementary Properties of Rings

5.1 Basic Properties

#	Property	Statement
1.	$a \cdot 0 = 0 \cdot a = 0$	Using distributivity
2.	$a(-b) = (-a)b = -(ab)$	Additive inverse property
3.	$(-a)(-b) = ab$	Product of two negatives
4.	$a(b-c) = ab - ac$	Distributivity
5.	If ring has unity, $(-1)a = -a$	Unity property
6.	If ring has unity, $(-1)(-1) = 1$	Unity property
7.	Additive identity 0 and additive inverse $-a$ are unique	Group property
8.	Multiplicative identity (if exists) is unique	Ring property

Zero Property: $a \cdot 0 = 0 \cdot a = 0$

5.2 Zero Divisors

Definition: In a ring R , non-zero elements a and b are called **zero divisors** if:

$$ab = 0 \text{ with } a \neq 0, b \neq 0$$

Example: In $\mathbb{Z}_6$, $2 \cdot 3 = 6 \equiv 0$, so 2 and 3 are zero divisors.

5.3 Idempotent Element

Definition: An element $a \in R$ is called **idempotent** if:

$$a^2 = a$$

Examples:

- 0 and 1 are idempotent in any ring
- In Boolean rings, every element is idempotent

5.4 Units

Definition: In a ring with unity $1 \neq 0$, an element $u \in R$ is called a **unit** if it has a multiplicative inverse.

$$uu^{-1} = u^{-1}u = 1$$

Examples:

- In $\mathbb{Z}$, units are 1 and -1
- In $\mathbb{Z}_5$, units are 1, 2, 3, 4

Topic 6: Fields

6.1 Definition

A **field** is a commutative ring with unity $1 \neq 0$ in which **every non-zero element is a unit** (has a multiplicative inverse).

Field = Commutative Ring + Unity + Every non-zero element is a unit

6.2 Examples of Fields

Set	Description	Is it a Field?
$\mathbb{Q}$	Rational numbers	Yes
$\mathbb{R}$	Real numbers	Yes
$\mathbb{C}$	Complex numbers	Yes
$\mathbb{Z}$	Integers	No (only 1 and -1 are units)
$\mathbb{Z}_n$	Integers modulo n	If and only if n is prime

6.3 Why $\mathbb{Z}_p$ is a Field

When p is prime, $\mathbb{Z}_p$ is a field because:

1. $\mathbb{Z}_p$ is a commutative ring with unity
2. For any non-zero element $a \in \mathbb{Z}_p$, $\gcd(a, p) = 1$
3. Therefore a has a multiplicative inverse modulo p

6.4 Ring vs Field Comparison

Property	Ring	Field
Commutative Addition	Yes	Yes
Additive Identity (0)	Yes	Yes
Additive Inverses	Yes	Yes
Associative Multiplication	Yes	Yes
Commutative Multiplication	Not required	Yes
Multiplicative Identity (1)	Not required	Yes
Multiplicative Inverses	Not required	Yes (for all non-zero elements)
Zero Divisors	May have	None

Unit-IV: Important Questions

Short Answer Questions (2–5 Marks)

1. Define a ring and state its axioms.
2. Define a commutative ring with an example.
3. What is a ring with unity? Give an example.
4. What is the ring of polynomials? Give an example.
5. Define a field with an example.
6. What is a zero divisor? Give an example.
7. What is a unit in a ring?
8. State the elementary properties of rings.
9. Why is $\mathbb{Z}$ not a field?
10. When is $\mathbb{Z}_n$ a field?

Long Answer Questions (10 Marks)

1. Define a ring. Explain different types of rings with examples.
2. What is a ring of polynomials? Prove that $R[x]$ is a ring.
3. Explain the ring of functions with examples.
4. State and prove the elementary properties of rings.
5. Define a field. Show that $\mathbb{Z}_p$ is a field iff p is prime.
6. What is the difference between a ring and a field? Explain with examples.

Objective Questions (1 Mark)

1. A ring is an abelian group under:
(a) Addition (b) Multiplication (c) Both (d) None

Answer: (a)

2. $\mathbb{Z}$ is:
(a) Field (b) Ring (c) Both (d) None

Answer: (b)

3. $\mathbb{R}$ is:
(a) Field (b) Ring (c) Both (d) None

Answer: (c)

4. The ring of polynomials over $\mathbb{R}$ is denoted by:

- (a) $\mathbb{R}(x)$ (b) $\mathbb{R}[x]$ (c) Both (d) None

Answer: (b)

5. Which is not a field?

- (a) $\mathbb{Q}$ (b) $\mathbb{R}$ (c) $\mathbb{Z}$ (d) $\mathbb{C}$

Answer: (c)

Quick Revision

Ring Properties Summary

Component	Properties
($\mathbb{R}, +$) is an Abelian Group	Closure, Associativity, Commutativity, Identity (0), Inverses ($-a$)
Multiplication	Closure, Associativity
Distributivity	$a(b+c) = ab+ac$, $(a+b)c = ac+bc$
Optional Properties	Commutative: $ab=ba$, With Unity: has 1, Field: Commutative + Unity + All non-zero are units

Examples Summary

Structure	Examples
Commutative Ring	$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
Ring without Unity	$n\mathbb{Z}$
Non-commutative Ring	$M_{n \times n}(\mathbb{R})$
Ring of Polynomials	$\mathbb{Z}[x], \mathbb{Q}[x], \mathbb{R}[x]$
Ring of Functions	Set of functions $X \rightarrow R$

Field	$\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_p$ (p prime)
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Key Formulas

Property	Formula
Zero Property	$a \cdot 0 = 0 \cdot a = 0$
Negative Product	$a(-b) = (-a)b = -(ab)$
Product of Negatives	$(-a)(-b) = ab$
Distributivity	$a(b+c) = ab + ac$

End of Unit-IV Notes